Machine Learning Introduction – 2

EE698V - Machine Learning for Signal Processing

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Machine Learning

- Training set
  - Inputs, $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \dots\}$
  - Targets, $\{\boldsymbol{y}_1, \boldsymbol{y}_2, \dots\}$
- Test set
  - Inputs, $\{\boldsymbol{x}_1^t, \boldsymbol{x}_2^t, \dots\}$
  - Unseen targets, $\{\boldsymbol{y}_1^t, \boldsymbol{y}_2^t, \dots\}$

ground truth

$x_t \rightarrow \boxed{F} \rightarrow y_t$

(h, d)

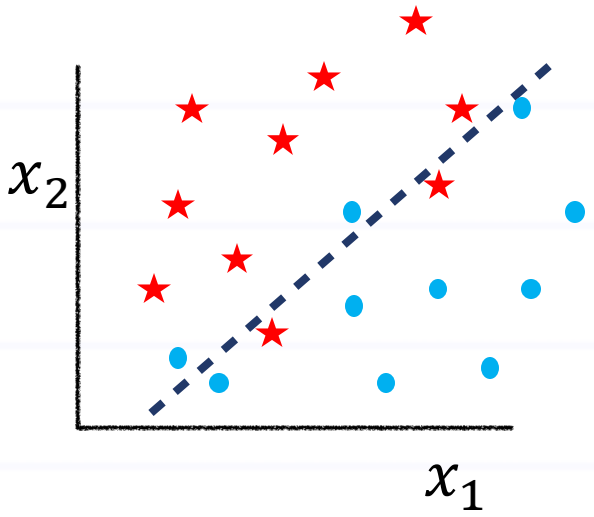
$y \in \{R, L, J\}$

x_1 h	dia	$\hat{y}$ or t
5.1	3.0	L
4.8	2.5	L
2.5	5.2	R
2.6	2.3	J

Evaluation Measures for Classification

Confusion Matrix

		PREDICTED	
		NEGATIVE	POSITIVE
ACTUAL	NEGATIVE	True Neg.	False Pos.
	POSITIVE	False Neg.	True Pos.



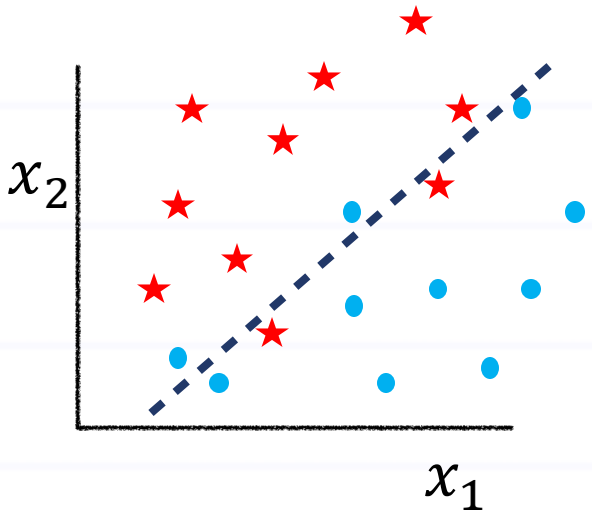
300

g.t. \ pred.	R	L	J
R	20	60	20
L	15	80	5
J	20	30	50

gt	pred
R	R
R	J
L	R

Accuracy

- $Accuracy = \frac{correct}{all} = \frac{TP+TN}{TP+TN+FP+FN}$



$A = \frac{150}{300} = 50\%$

Problems with Accuracy

- E.g.
  - actually 990 R, 10 L
  - predicted all R

gt \ p	×	✓
×	960	30
✓	5	5

$A = 0.965$

	×	✓
×	990	0
✓	10	0

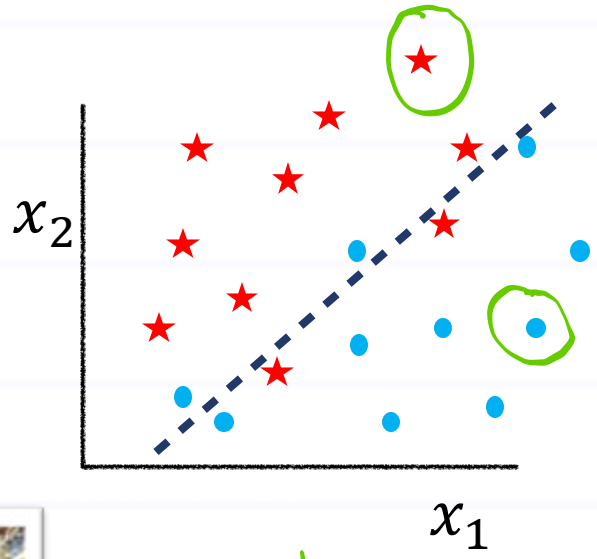
$A = 0.99$

$R = \frac{5}{10}$ $\quad \frac{0}{10}$

$P = \frac{5}{35}$ $\quad \frac{0}{0}$

Precision and Recall

- $P = \frac{TP}{TP+FP}$ $= \frac{8}{10}$



- $R = \frac{TP}{TP+FN}$ $= \frac{8}{10}$

gt. \ pred.	×	✓
×	8	2
✓	2	8

precision

recall

F-score

- $F = harmonic\ mean(P, R) = \left(\frac{P^{-1}+R^{-1}}{2}\right)^{-1} = \frac{2PR}{P+R}$

P	R	F
↑	↑	↑
↑	↓	↓
↓	↑	↓
↓	↓	↓

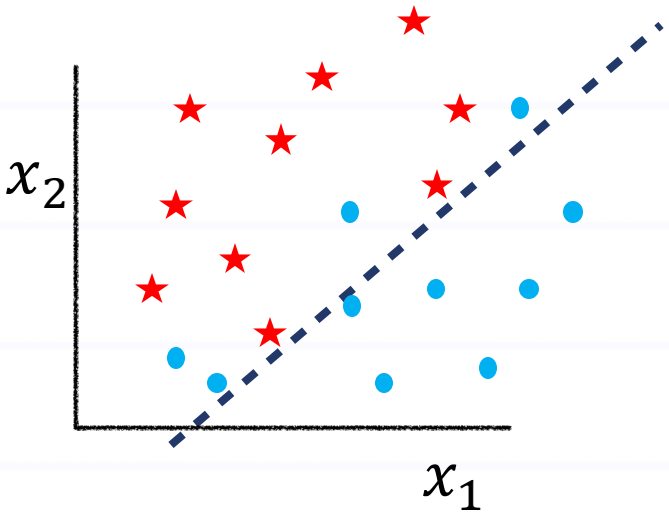
$$F = \frac{1 \times 1 \times 2}{1+1} = 1$$

$$F = \frac{1 \times 0.1 \times 2}{1.1} = \frac{0.2}{1.1} \sim .18$$

$$F = \frac{0.1 \times 0.1 \times 2}{.1+.1} = 0.1$$

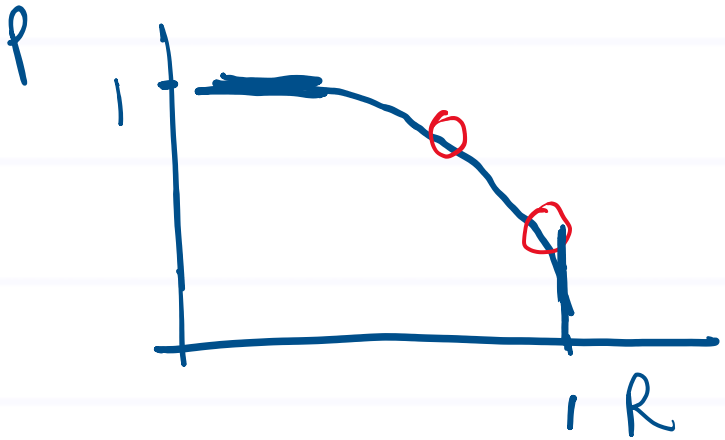
PRC curve

- Change in Precision vs Recall w.r.t. a threshold, e.g. c



	×	✓
×	6	4
✓	0	10

P	R
0.8	0.8
$\frac{10}{14}$	1
1	0.1
$\frac{10}{19}$	1



Practice problems

- Give some example tasks where we need high precision, but recall can be somewhat compromised
- Give some example tasks where we need high recall, but precision can be somewhat compromised

Typical Steps

1. Fix Goal

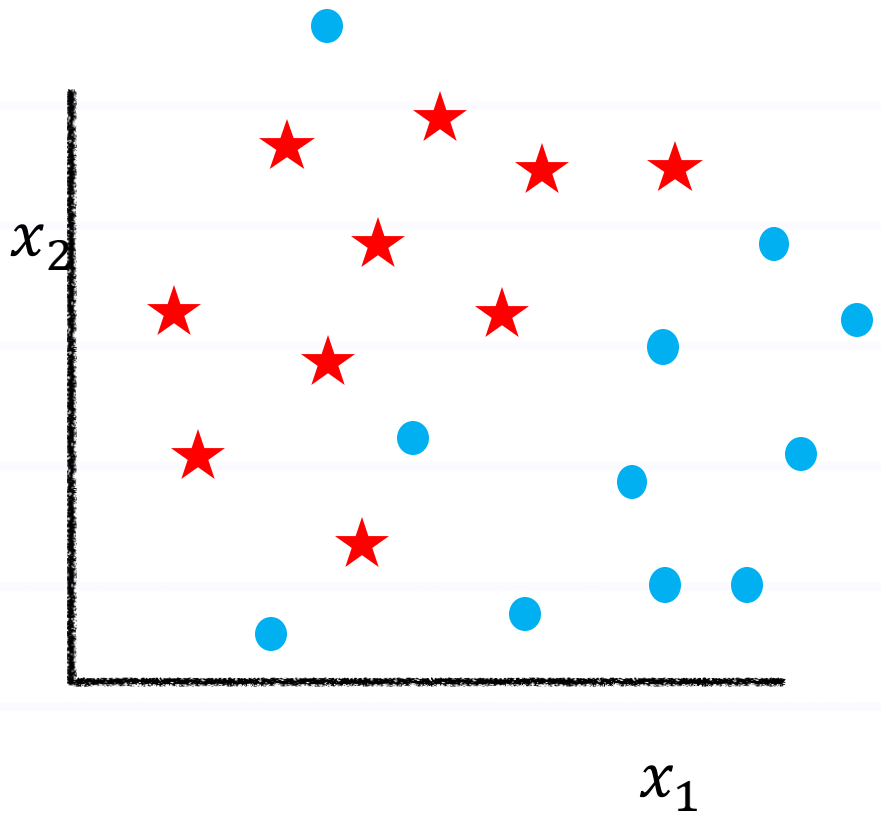
- Classification of flowers into 3 classes

2. Data Collection

- Collect relevant data
- Data variability – select scope of generalization
- Least effort desirable

2. Data Collection

- Noise: errors in measurement due
  - measuring device
  - background phenomena
  - data size artifacts
  - …
- Outlier: data point that differs significantly from the normal trend



3. Data Annotation/Labeling

- Data denoising
- Make conventions for data labeling
  - boundary cases
  - background noise
- Train the team of annotators
- Continuous annotation
- Monitor Annotation Quality
  - Re-train annotators
  - Tune annotation conventions

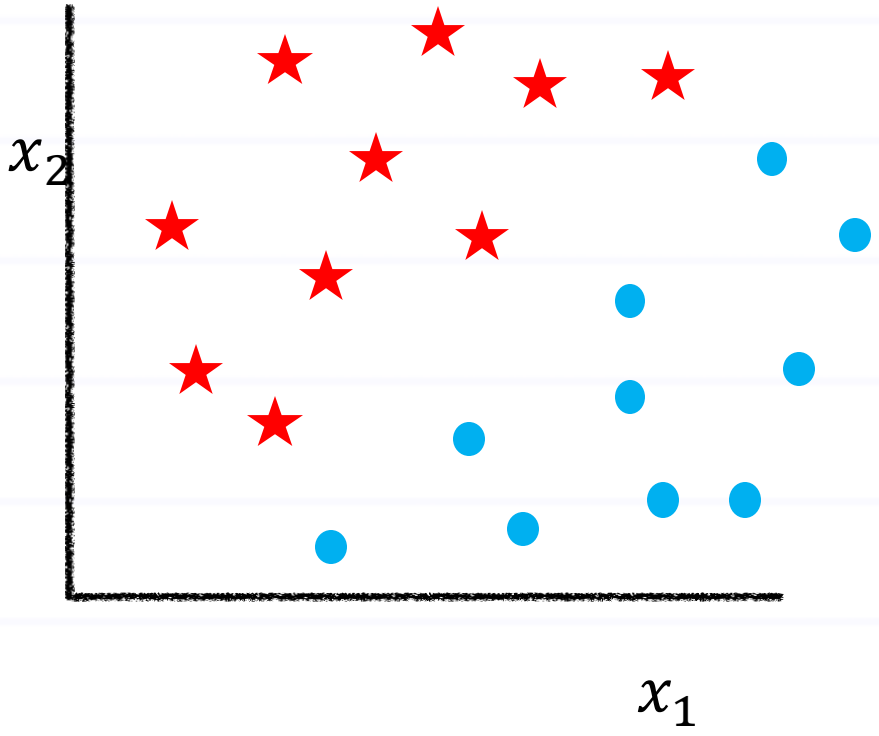
4. Feature Design/Extraction

(height, diameter)

- Find relevant features
  - Informative
  - Invariant
  - Compact

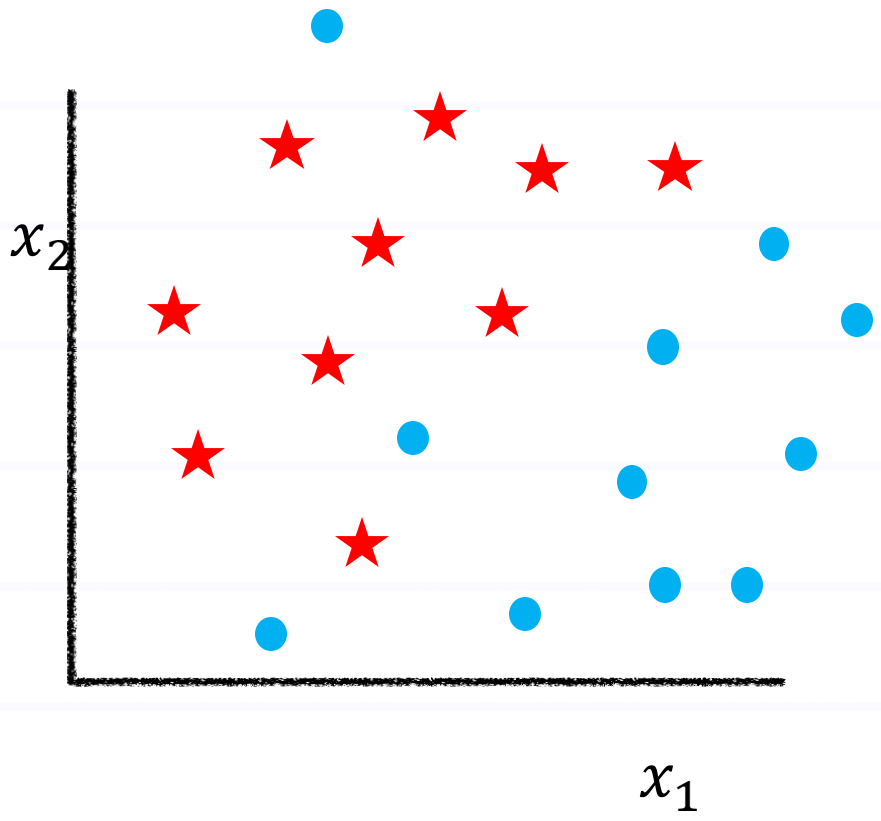
5. Model Selection and Training

- Select what model to use
  - high performance
  - computationally efficient
- Use training data to find optimal values of model parameters (m,c)



6. Handling noise and outliers

- Overfitting
- Validation



7. Testing

- Performance on test data
- Variability

8. Deployment

- Release to beta users
- Release in market

9. Debugging and Model Improvement

- Gather more data for training
- Error analysis
- Go back to step 3

References

- "Pattern Recognition and Machine Learning", C.M. Bishop, 2nd Edition, Springer, 2011.
  - **Section 1.0**